

WASP-36b

R-1736

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_190325 · created 2026-08-02T18:06:27+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458567.7029 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.014
Transit depth (Rp/Rs)^2	2.27 +/- 0.41 [%]
Orbital Inclination (inc)	88.08 +/- 1.19
Airmass coefficient 1 (a1)	0.952 +/- 0.014
Airmass coefficient 2 (a2)	0.033 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.48 %
Best Comparison Star	#2 - [426, 127]
Optimal Aperture	2.9
Optimal Annulus	11.6
Transit Duration (day)	0.0949 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.014 vs 0.1368 (deviation 0.93σ)
Duration fit vs published	0.0949 d vs 0.0758 d (deviation 7.93σ)
Mid-time O–C	3.0 min
Depth SNR	9.9
Residual scatter	1.48 %

Light curve

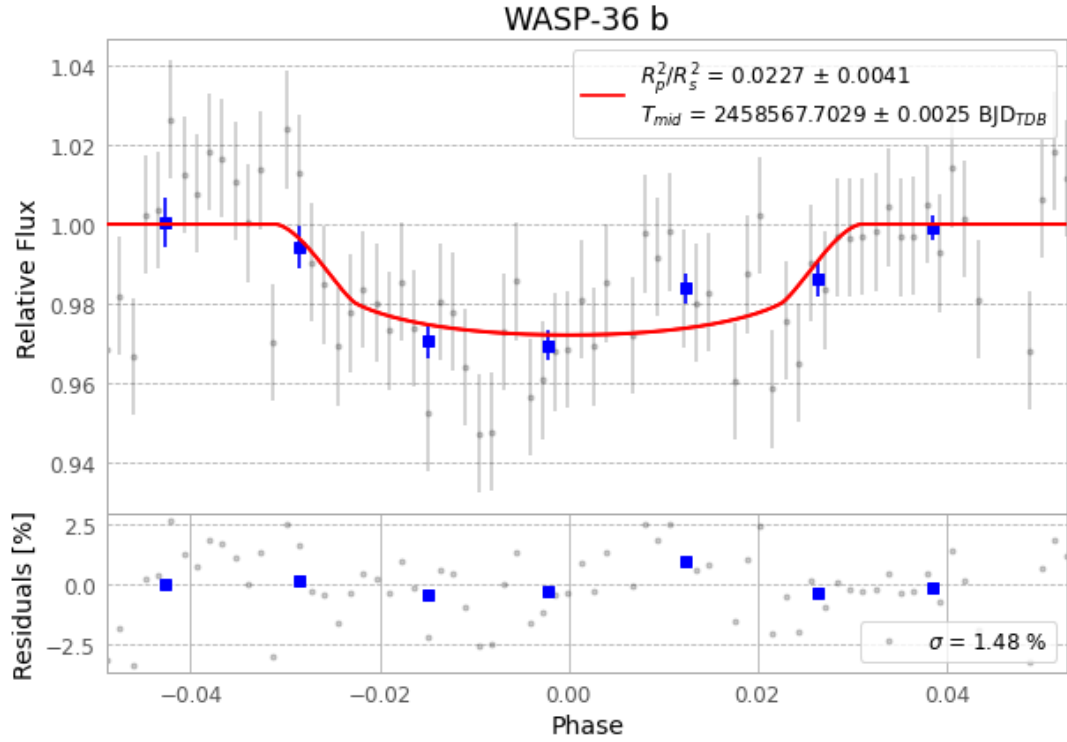


Figure 1 — FinalLightCurve_WASP-36 b_2019-03-24.png (SHA-256 341527b264f06656...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 159], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 647eca467d80dbc1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-36190325025712.FITS → WASP-36190325064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-190325.log (67605ba233c2...), FinalLightCurve_WASP-36 b_2019-03-24.png (341527b264f0...), FinalLightCurve_WASP-36 b_2019-03-24.csv (f6957396e56d...)

Compute resources

Wall time 205.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:06Z.